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Leveraging Class Distributions in CLIP for Weakly Supervised Semantic Segmentation

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Abstract

Image-level Weakly Supervised Semantic Segmentation (WSSS) typically leverages Class Activation Maps (CAMs) for pixel-wise localization. However, existing CLIP-based methods often yield under-activated CAMs, primarily due to the inaccurate semantic relationships in the affinity-based refinement. In this work, we propose a novel framework, CD-CLIP (Class Distribution based CLIP), which addresses this issue by introducing a Class Distribution Aware (CDA) module. The CDA module captures richer semantic relationships by modeling patch-wise distributions across all classes using Jensen-Shannon divergence, thereby enhancing the completeness of CAMs. While this significantly improves the coverage of the foreground class, the over-activation at class boundaries might also exist due to the comprehensive integration of relationships between inter target classes. To mitigate this adverse effect on segmentation supervision, we introduce a Super-class Boundary Exploration (SBE) module, which leverages structural knowledge of DINO to generate boundary-aware super-class prototype CAMs. By employing the boundary-enhanced loss, our SBE module effectively provides accurate boundary supervision for the final segmentation. Our proposed CD-CLIP framework achieves state-of-the-art performance on both PASCAL VOC and MS COCO benchmarks. Code is available here.

1. Introduction

Weakly Supervised Semantic Segmentation (WSSS) has been introduced in recent years to generate pixel-level predictions using only weak annotations. It effectively reduces the expensive annotation cost and labor consumption compared with the fully supervised semantic segmentation relying on pixel-wise annotations. The weak annotations include points [3], scribbles [26, 41], bounding boxes [9, 23], and image-level labels [1, 31, 52]. We focus on the popular but challenging image-level labels in this paper.

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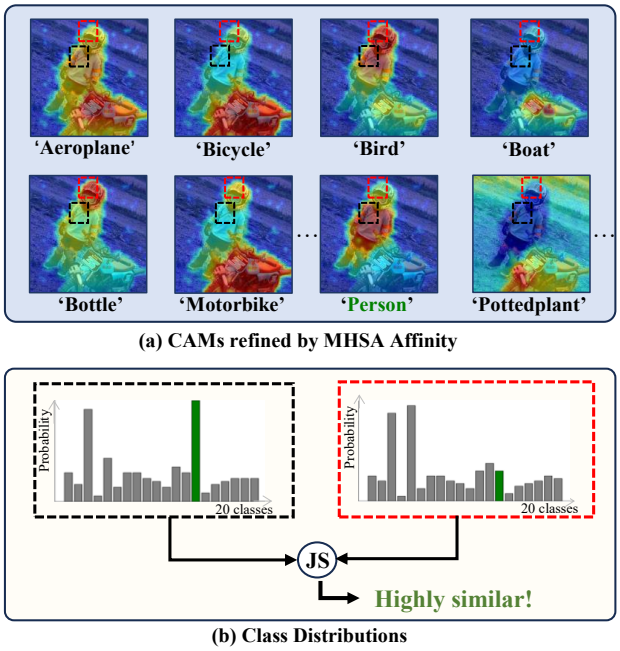


Figure 1. Our motivation. (a) CAMs from all classes refined by MHSA affinity in CLIP. The selected patches in the same target class are marked in 'red box' and 'black box'. The MHSA affinity fails to establish the correct relationships between these two selected patches. (b) Class distributions for two selected patches. We measure the Jensen-Shannon (JS) divergence between class distributions of two patches. The results indicate that patches within the same target class exhibit highly similar class distributions, motivating us to identify the correct relationship through these distributions.

In image-level WSSS, Class Activation Maps (CAMs), renowned for their localization capabilities, are widely used to generate pseudo-labels, which then provide dense supervision for the segmentation decoder. Recent advancements have explored leveraging Contrastive Language-Image Pre-training (CLIP), pre-trained on 400 million image-text pairs, for CAM generation. However, due to the limited semantic information derived from image-level labels, the initial CAMs produced by CLIP without a dedicated train-

ing process are typically coarse and primarily activate only the most distinctive regions, impacting WSSS performance.

To address this limitation, existing CLIP-based approaches typically employ Multi-Head Self-Attention (MHSA) weights from Vision Transformer (ViT) encoders to model patch-wise semantic affinity, where spatially proximate patches with similar semantics are assigned strong relationships. Although this affinity-based refinement can expand CAM activation regions, we identify the critical shortcoming: as shown in Figure 1(a), the derived affinity often fails to establish accurate intra-class semantic relationships, resulting in the under-activation of target class (e.g., the 'head' remaining inactive in 'person'). Additionally, in Figure 1(b), the class distribution analysis reveals an intriguing phenomenon: while these selected foreground patches show divergent responses to target class (e.g., high 'person' score for black patch vs. low score for red patch), they exhibit highly similar class distribution across certain irrelevant classes. This discovery suggests that the rich class distribution, rather than just target-class activations, contains more valuable semantic information. By leveraging this distributional knowledge across all classes, we can achieve more precise and comprehensive CAM refinement, effectively addressing the under-activation problem in current affinity-based approaches.

Inspired by our aforementioned finding, we propose a Class Distribution Aware (CDA) module to leverage the class distribution knowledge for building precise relationships in the affinity, leading to more complete CAMs. Specifically, we first generate the attention maps between image patches and text features from CLIP, and then, for each patch in the attention map, we estimate its specific class distribution for both target classes and super-classes (designed for enriching the distribution representation). Next, we construct the similarity map of class distribution between two patches and all other patches in the image through the Jensen-Shannon (JS) divergence. As patches from the same target class show highly similar class distributions, we rectify the initial affinity from MHSA by selectively incorporating these constructed similarities and dissimilarities, and obtain the Class Distribution Aware (CDA) affinity. This CDA affinity can be effectively used in WSSS to enhance the CAM capability of accurately localizing class-relevant patches without requiring further training.

CDA affinity excels at activating a more complete CAM, yet the highly similar class distribution at boundary regions between inter target classes ('motorbike' and 'person' in Figure 1(a)) might also inadvertently induce the strong relationships. This inevitably leads to over-activation in CAM boundary regions, resulting in inaccurate boundary supervision. To mitigate that effect, we also propose a Super-class Boundary Exploration (SBE) module. This module provides dedicated boundary supervision among super-classes for final segmentation prediction, effectively addressing the over-activation boundary issue unaddressed by the CDA module. In detail, it incorporates a super-prototype CAM, leveraging DINO (Distillation with No Labels) image features, and a super-class CAM, inferred through the zero-shot ability of CLIP. As image features of DINO are known to provide better separation of super-class regions [17], this integration results in more accurate and compact boundary delineation. Therefore, we employ this super-prototype CAM to specifically supervise the boundary segmentation prediction among different super-classes through a boundary-enhanced loss. By integrating our two proposed modules, we establish a new effective framework CD-CLIP for single-stage image-level WSSS.

The contributions of this paper are summarized as follows:

- Our analysis reveals that while conventional MHSA-based affinity fails to establish reliable intra-class relationships, the distribution across all classes captures richer and more discriminative semantic relationships among various patches.
- Based on our findings, we propose a CD-CLIP framework, which consists of a Class Distribution Aware (CDA) module to achieve a more complete CAM by leveraging class distributions over all classes for each image patch, and a Super-class Boundary Exploration (SBE) module to provide more accurate boundary supervision for the final segmentation prediction.
- The experimental results demonstrate that our CD-CLIP framework outperforms both other single-stage and multi-stage methods, achieving 82.5% mIoU on PASCAL VOC 2012 val set and 54.1% mIoU MS COCO val set.

2. Related Work

2.1. Weakly Supervised Semantic Segmentation

Weakly Supervised Semantic Segmentation (WSSS) [21, 32, 42, 43] using only image-level annotations has garnered significant attention in recent years. Solutions to this task generally fall into two primary categories: multi-stage and single-stage frameworks. In the multi-stage framework [16, 20, 60], the process typically involves sequentially training a classification network followed by a segmentation network. The classification network first generates initial Class Activation Maps (CAMs). These CAMs are then refined using post-processing techniques like IRN [2] or PSA [1] to obtain pseudo-labels. Subsequently, the segmentation network is trained to predict segmentation results under the supervision of these pseudo-labels. Additionally, to address the under-activation issue, prototype-based methods [4, 6, 11, 45, 56] aim to improve prototype representations to expand these activation regions within CAMs.

The single-stage framework [13, 35], in contrast, adopts

a parallel training paradigm for both the classification and segmentation networks. Specifically, ViT-based methods frequently align with the ToCo [35] framework, employing a patch token contrast loss to mitigate the common over-smoothing problem. DuPL [46] tackles the inherent confirmation bias of pseudo-labels by developing two subnets that provide cross-supervision for the erroneous regions within these pseudo-labels. Moreover, WeCLIP [54] proposes a robust frozen backbone specifically tailored for single-stage CLIP-based methodologies, alongside a novel CLIP feature decoder for segmentation prediction [30]. Deviating from these prior works, our method integrates an additional DINO image feature to produce super-prototype CAMs, which serve as super-class boundary supervision.

2.2. Affinity Learning for WSSS

Affinity, conceptualized as a matrix capturing similarity relationships between patches (or pixels), has found extensive application in the Weakly Supervised Semantic Segmentation (WSSS) task for bolstering Class Activation Map (CAM) performance. Notably, Ahn et al. [1] introduced AffinityNet, a Deep Neural Network (DNN) trained to predict semantic affinity between adjacent image coordinates, supervised by initial discriminative part segmentations. Given the substantial performance gains achieved by Transformers in WSSS, the affinity inherent in Multi-Head Self-Attention (MHSA) has been investigated for refining initial pseudo-labels through the AFA module [34]. Different from these prior methods, CLIP-ES utilizes CAMs as class masks to derive class-wise affinity without necessitating any additional training. Our work, building upon these advancements, capitalizes on class distributions present in the CLIP model to unveil more precise relationships for subsequent CAM refinement.

3. Methodology

This section introduces CD-CLIP, our single-stage framework for weakly supervised semantic segmentation (WSSS). As illustrated in Figure 2, the pipeline first generates initial Class Activation Maps (CAMs) and affinity, then employs a Class Distribution Aware (CDA) module to refine CAMs using CLIP's class distribution-based relationships in affinity. Finally, a Super-class Boundary Exploration (SBE) module mitigates boundary imperfections caused by these distribution-based relationships by providing more accurate boundary supervision.

3.1. CAM and Affinity Generation

CAM is frequently used in image-level WSSS for identifying discriminative regions in the image due to its localization ability, which are then further refined into pseudo labels as pixel-level supervision. In our method, we follow GradCAM [27] to generate the initial CAM. Formally, given an image $\mathbf{X} \in \mathbb{R}^{3\times H\times W}$ and text prompts of C classes, we output image features $\mathbf{F} \in \mathbb{R}^{d\times h\times w}$ and text features $\mathbf{T} \in \mathbb{R}^{C\times d}$ from image encoder and the text encoder separately, where d is feature dimension, h and w are spatial sizes. Then, the class logits are calculated by the distance between image features and text features, and the initial CAM $\mathbf{M}^c_{init}$ for class index c is formulated as:

$$\mathbf{M}^c_{init} = \mathbf{ReLU}(\sum_k \mathbf{W}^c_k \mathbf{V}^k), \tag{1}$$

where $\mathbf{W}^c_k$ represents the weight in feature map k for class c, derived from the gradient between class logits and feature maps. $\mathbf{V}^k$ is the activation value for feature map k, and $\mathbf{ReLU}(\cdot)$ function is applied to eliminate negative values. Moreover, we design a super-class set $\mathbf{D}_\mathrm{s}$ containing N super-classes (*e.g.*, 'vehicle' and 'animal') for target classes, and compile their class information through the text encoder. By doing so, we not only use super-class information to provide more abundant class distributions for our CDA module, but also infer super-class CAMs $\mathbf{M}^n_s$ via Eq. (1) to serve as guidance for our subsequent SBE module.

Meanwhile, we also extract all attention weights $\mathbf{A} \in \mathbb{R}^{hw\times hw}$ from the MHSA in image encoder to calculate initial affinity $\mathbf{A}_{init}$ as follows:

$$\mathbf{A}_{init} = \frac{\mathbf{Sinkhorn}(\mathbf{A}) + \mathbf{Sinkhorn}(\mathbf{A})^\mathrm{T}}{2}. \tag{2}$$

where $\mathbf{Sinkhorn}(\cdot)$ denotes the Sinkhorn normalization [36]. $\mathbf{A}_{init}$ captures long-range semantic relationships between image patches. However, as shown in Figure 1(a), these semantic relationships in $\mathbf{A}_{init}$ tend to be inaccurate and fail to activate all regions of the target class, making them suboptimal for localization in CAM refinement.

3.2. Class Distribution Aware (CDA)

As explained in Section 1, complete class distributions [18] in CLIP contain valuable semantic information, which can build more precise and comprehensive patch-wise relationships. Hence, our CDA module is designed to leverage this class distribution knowledge for building the stronger affinity, effectively addressing the under-activation problem on CAMs refined by initial affinity $\mathbf{A}_{init}$.

In detail, we first generate the attention maps $\mathbf{S_a}$ between image features $\mathbf{F}$ and all text features $\mathbf{T}_{all} \in \mathbb{R}^{|C+N|\times d}$ (target classes and designed super-classes), as follows:

$$\mathbf{S_a} = \mathbf{Min\text{-}max}(\mathbf{CosSim}(\mathbf{F}, \mathbf{T}_{all})), \tag{3}$$

where $\mathbf{Min\text{-}max}(\cdot)$ is the Min-max normalization and $\mathbf{CosSim}(\cdot)$ is the cosine similarity function. To enrich class distribution representation, we also incorporate super-class information. For each patch in the attention map $\mathbf{S_a}$,

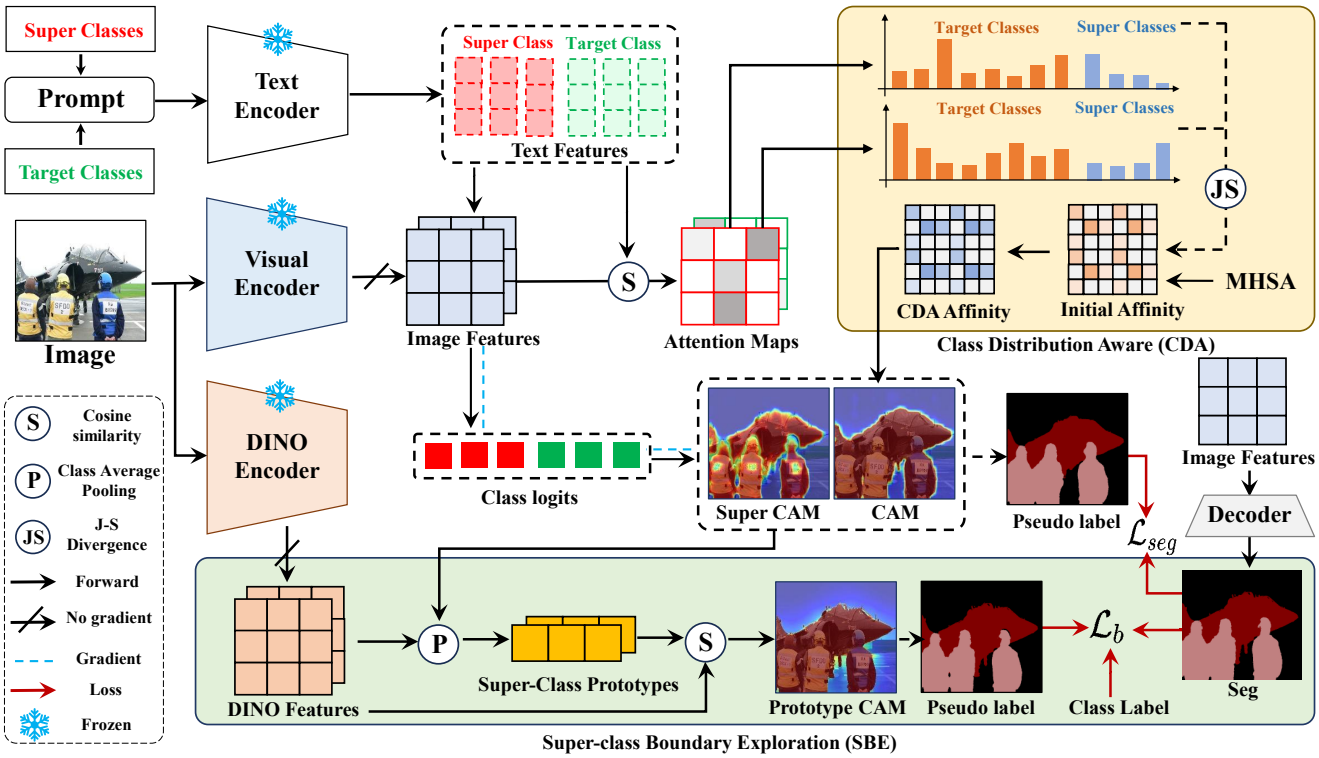


Figure 2. An overview of our framework CD-CLIP. The image is first fed into the CLIP image encoder and the DINO encoder to extract the corresponding features. Simultaneously, text prompts comprising target classes and designed super-classes are input into the CLIP text encoder to obtain text features. These image and text features are then used to generate attention maps based on their similarity. In the CAM generation process, we infer two kinds of CAMs based on the probability between image features and text features. In the Class Distribution Aware (CDA) module, we introduce CDA affinity for enhancing CAM refinement. This affinity is generated from JS divergence between class distributions within the attention maps and the initial affinity captured by Multi-Head Self-Attention (MHSA). In the Super-class Boundary Exploration (SBE) module, we derive a Super-class prototype CAM, providing accurate boundary supervision via a boundary-enhanced loss. This prototype CAM is generated using DINO features and the inferred super-class CAM. For the decoder, it outputs the final segmentation predictions, which are supervised by both pseudo labels derived from refined CAMs and Super-class prototype CAMs.

it carries a distribution of scores corresponding to various class semantics. Then, the class distribution similarity map $\mathbf{S_d}$ can be defined as:

$$\mathbf{S_d} = \frac{(1 - \mathbf{D_{JS}}(\mathbf{Softmax}(\mathbf{S_a})||\mathbf{Softmax}(\mathbf{S_a})^{\mathrm{T}}))}{\tau}, \quad (4)$$

where $\mathbf{Softmax}(\cdot)$ is used for estimating the class distribution probability for all patches in $\mathbf{S_a}$. $\mathbf{D_{JS}}(\cdot||\cdot)$ is the Jensen-Shannon (JS) divergence, which is a finite and symmetric metric, measuring pairwise similarity. The higher similarity between two class distributions corresponds to the smaller JS divergence. τ is the temperature to control the softness.

As patches sharing the same class semantic exhibit similar class distributions, we aim to leverage this distribution-based similarity in $\mathbf{S_d}$ to rectify the initial affinity. Specifically, by highlighting the relationships among patches within the same class, and disregarding those between patches within different classes, we obtain our Class Distribution Aware (CDA) affinity $\mathbf{A}_d$ as follows:

$$\mathbf{A}_d = \mathbf{Softmax}(\mathbf{S_d}) \cdot \mathbf{A}_{init}, \quad (5)$$

where all inaccurate relationships in the initial affinity $\mathbf{A}_{init}$ are rectified by distribution-based similarity $\mathbf{S_d}$.

After that, this CDA affinity can be used to refine the CAMs precisely through the following process:

$$\mathbf{M}^c_{re} = \mathbf{M}^c_{init} \cdot (\mathbf{A}_d)^t. \quad (6)$$

where $\mathbf{M}^c_{re}$ is the refined CAM, and t is the refining iterations. During this process, patches with strong relationships tend to share the same class semantics and become activated on CAMs.

In this way, our CDA module harnesses class distribution relationships among patches to produce precise and robust CDA affinities. This affinity is instrumental in solving the

under-activation issue and improving the localization capability during CAM refinement, thereby enabling the derivation of high-quality pseudo labels for effective supervision.

3.3. Super-class Boundary Exploration (SBE)

While our CDA module can build precise intra-class relationships, it might remain a persistent gap in the establishment of strong relationships between inter target classes. This may lead to over-activation at the CAM boundary, resulting in inaccurate boundary supervision. To resolve this unaddressed issue of CDA module, we introduce a complementary Super-class Boundary Exploration (SBE) module.

Unlike conventional WSSS features, DINO features exhibit superior representational capacity for super-class region segmentation [17]. Therefore, our SBE module combines DINO's segmentation strengths with class-aware activation to capture complete super-class regions and precise boundaries by the generation of super-class prototype CAMs. This super-class prototype CAM is further used to supervise the final segmentation prediction, mitigating the over-activation effects.

However, DINO features cannot provide class-specific localization for the WSSS task. To solve this, we first utilize the inferred super-class CAMs $\mathbf{M}_s^n$ from the CLIP model as the class-specific mask and calculate the super-class prototypes based on the DINO image features $\mathbf{F_d} \in \mathbb{R}^{d\times h\times w}$, the process is defined as:

$$\mathbf{f}_s^n = \mathbf{CAP}\left(\mathbf{M}_s^n \odot \mathbf{F}_d\right), \tag{7}$$

where $\mathbf{f}_s^n$ is the super-class prototype for each super-class, and $\mathbf{CAP}(\cdot)$ denotes the class average pooling operation.

Next, recognizing that super-class prototypes effectively capture their respective super-class semantics, we generate the super-class prototype CAMs $\mathbf{M}_p^n$ by computing the similarity between $\mathbf{f}_s^n$ and each patch in the DINO image features $\mathbf{F}_d$:

$$\mathbf{M}_p^n = \mathbf{ReLU}(\mathbf{CosSim}(\mathbf{f}_s^n, \mathbf{F}_d)), \tag{8}$$

Compared with the initial CAMs $\mathbf{M}_{init}^c$, the super-class prototype CAMs $\mathbf{M}_p^n$ with DINO feature knowledge include more accurate boundary activation between super-classes.

To ensure accurate supervision while minimizing computational cost, we focus on supervising segmentation predictions that contain boundaries between super-classes. To do this, we should identify the super-classes present in each segmentation prediction from the decoder. Specifically, we leverage the target class label and the designed super-class set $\mathbf{D}_\mathrm{s}$ to determine the presented super-classes. For each prediction, we check the number of presented classes based on the class label. If multiple classes are present, we map them to their corresponding super-classes from the set (*e.g.*, 'cat' belongs to 'animal'). This allows us to obtain the set of presented super-classes $\mathbf{C}_\mathrm{p}$ for each prediction.

We then use $\mathbf{M}_p^n$ to derive the pseudo label $\mathbf{Y}_s$. For predictions with multiple super-classes, we provide super-class boundary supervision through our designed boundary-enhanced loss $\mathcal{L}_b$, which is computed as follows:

$$\mathcal{L}_b = \begin{cases} \mathbf{Dice}(\mathbf{P}, \mathbf{Y}_s) & \text{if} \quad |\mathbf{C}_\mathrm{p}| \geq 2, \\ 0, \text{otherwise.} \end{cases} \tag{9}$$

where $\mathbf{P}$ is the segmentation prediction, $\mathbf{Dice}(\cdot,\cdot)$ represents the Dice loss, and $|\mathbf{C}_\mathrm{p}|$ represents the number of presented super-classes for the current prediction $\mathbf{P}$. By employing $\mathcal{L}_b$, we effectively supervise boundary predictions between super-classes using image knowledge from DINO. This encourages the decoder to produce more complete and reliable segmentation predictions by avoiding over-activated boundary regions.

3.4. Training objective

As illustrated in Figure 2, our framework encompasses two training objectives: the segmentation loss $\mathcal{L}_{seg}$ and the boundary-enhanced loss $\mathcal{L}_b$. Both of them provide effective supervision for the final segmentation prediction. The overall training objective of our framework is defined as follows:

$$\mathcal{L} = \mathcal{L}_{seg} + \lambda_1 \mathcal{L}_b. \tag{10}$$

where λ_1 is the weight factor.

4. Experiments

4.1. Experimental Settings

Datasets and Evaluation Metrics Following common image-level WSSS practice [55], we evaluated our framework on two benchmarks: PASCAL VOC 2012 and MS COCO 2014. The PASCAL VOC 2012 dataset, which has 21 classes, is augmented with SBD [12] and comprises 10,582 images for training, 1,449 for validation, and 1,456 for testing. The MS COCO 2014 dataset features 81 classes, with 82,081 images for training and 40,137 for validation. We use mean Intersection over Union (mIoU) as our primary evaluation metric.

Implementation Details We adopt the frozen CLIP with ViT-Base-16 architecture [10] as our visual encoder and the frozen DINOv2-ViT-Base-14 architecture [57] as DINO encoder. For the decoder, we use a light transformer-based architecture, as designed in WeCLIP. During training, we optimize the entire network using AdamW [28] with its default parameters. All images are cropped to 320 $\times$ 320 for the visual encoder and 308 $\times$ 308 for the DINO encoder. The weighting factor λ_1 is set to 0.6. For the PASCAL VOC 2012 dataset, we set the batch size to 4 and the maximum iterations to 30,000.

Table 1. Performance comparison with other methods on PASCAL VOC 2012 validation and test split. 'Sup.' denotes the supervision type. '$\mathcal{I}$' denotes using image-level supervision. '$\mathcal{S}$' denotes using saliency map supervision. '**L**' denotes using language supervision. 'Net.' denotes the semantic segmentation model of multi-stage WSSS methods or the backbone of single-stage WSSS methods. 'RN' denotes ResNet network, 'MiT' denotes mix transformer, 'ViT' denotes vision transformer.

Method	Sup.	Net.	Val	Test
Multi-stage WSSS methods				
L2G [15] *CVPR'22*	$\mathcal{I}+\mathcal{S}$	RN101	72.1	71.7
RCA [59] *CVPR'22*	$\mathcal{I}+\mathcal{S}$	RN38	72.2	72.8
SIPE [6] *CVPR'22*	$\mathcal{I}$	RN101	68.8	69.7
AMN [24] *CVPR'22*	$\mathcal{I}$	RN101	70.7	70.6
CLIMS [47] *CVPR'22*	$\mathcal{I}+\mathbf{L}$	RN101	70.4	70.0
MCTformer [48] *CVPR'22*	$\mathcal{I}$	RN38	71.9	71.6
ESOL [25] *NeurIPS'22*	$\mathcal{I}$	RN101	69.9	69.3
OCR [8] *CVPR'23*	$\mathcal{I}$	RN38	72.7	72.0
BECO [33] *CVPR'23*	$\mathcal{I}$	MiT-B2	73.7	73.5
CLIP-ES [27] *CVPR'23*	$\mathcal{I}+\mathbf{L}$	RN101	73.8	73.9
CPAL [37] *CVPR'24*	$\mathcal{I}+\mathbf{L}$	RN101	74.5	74.7
PSDPM [58] *CVPR'24*	$\mathcal{I}+\mathbf{L}$	RN101	74.1	74.9
CTI [53] *CVPR'24*	$\mathcal{I}$	RN38	74.1	73.2
S2C [19] *CVPR'24*	$\mathcal{I}$	SAM	78.2	77.5
Single-stage WSSS methods				
ToCo [35] *CVPR'23*	$\mathcal{I}$	ViT-B	71.1	72.2
PFSR [13] *AAAI'24*	$\mathcal{I}$	ViT-B	75.7	75.0
SeCo [49] *CVPR'24*	$\mathcal{I}$	ViT-B	74.0	73.8
DuPL [46] *CVPR'24*	$\mathcal{I}$	ViT-B	73.3	72.8
WeCLIP [54] *CVPR'24*	$\mathcal{I}+\mathbf{L}$	ViT-B	76.4	77.2
DALNet [14] *ECCV'24*	$\mathcal{I}+\mathbf{L}$	ViT-B	74.5	74.9
FFR [51] *CVPR'25*	$\mathcal{I}$	ViT-B	76.0	75.5
ExCEL [50] *CVPR'25*	$\mathcal{I}+\mathbf{L}$	ViT-B	78.4	78.5
Ours	$\mathcal{I}+\mathbf{L}$	ViT-B	**82.5**	**82.4**

The designed super-class set $\mathbf{D}_s$ for PASCAL VOC 2012 is $\{$animal, vehicle, household item, furniture, person$\}$. For the MS COCO 2014 dataset, the batch size is 8 and the maximum iterations are 80,000. The details of super-class set for COCO are listed in the appendix. The learning rate for both datasets is 2e-5 with a weight decay of 0.01. During the inference stage, we utilize the additional Dense CRF post-processing [5] and apply the multi-scale factors as $\{1.0, 1.5\}$.

4.2. Experimental Results

Segmentation results comparison In Table 1, we present a comparative analysis of our method's final segmentation results against other state-of-the-art methods on both the validation and test splits of PASCAL VOC 2012.

Table 2. Performance comparison with other methods on MS COCO 2014 validation split.

Method	Sup.	Net.	Val
Multi-stage WSSS methods			
RIB [22] *NeurIPS'21*	$\mathcal{I}+\mathcal{S}$	RN101	43.8
SIPE [6] *CVPR'22*	$\mathcal{I}$	RN101	40.6
AMN [24] *CVPR'22*	$\mathcal{I}$	RN101	44.7
MCTformer [48] *CVPR'22*	$\mathcal{I}$	RN38	42.0
ESOL [25] *NeurIPS'22*	$\mathcal{I}$	RN101	42.6
OCR [8] *CVPR'23*	$\mathcal{I}$	RN38	42.5
BECO [33] *CVPR'23*	$\mathcal{I}$	RN101	45.1
CTI [53] *CVPR'24*	$\mathcal{I}$	RN101	45.4
KTSE [7] *ECCV'24*	$\mathcal{I}$	RN38	45.9
Single-stage WSSS methods			
AFA [34] *CVPR'22*	$\mathcal{I}$	MiT-B1	38.9
ToCo [35] *CVPR'23*	$\mathcal{I}$	ViT-B	42.3
DuPL [46] *CVPR'24*	$\mathcal{I}$	ViT-B	44.6
SeCo [49] *CVPR'24*	$\mathcal{I}$	ViT-B	46.7
WeCLIP [54] *CVPR'24*	$\mathcal{I}+\mathbf{L}$	ViT-B	47.1
ExCEL [50] *CVPR'25*	$\mathcal{I}+\mathbf{L}$	ViT-B	50.3
Ours	$\mathcal{I}+\mathbf{L}$	ViT-B	**54.1**

Our proposed method demonstrably outperforms all prior approaches, establishing new state-of-the-art performance with 82.5% mIoU on the validation set and 82.4% mIoU on the test set. This represents a substantial improvement of 6.1% and 5.2% over our baseline method, WeCLIP [54]. Furthermore, our single-stage method significantly surpasses multi-stage approaches that rely on additional post-processing techniques. For instance, we exceed CPAL [37] by impressive margins of 8.0% and 7.7%, and outperform PSDPM [58] by 8.4% and 7.5%. Most notably, our method achieves significant improvements over all previous single-stage methods, particularly in comparison to other methods such as ExCEL [50] that utilizes large language model [29, 38–40, 44]. These results collectively underscore the robust effectiveness of our method.

In Table 2, our method continues to demonstrate superior segmentation performance on the MS COCO validation set, achieving the highest mIoU of 54.1% among existing state-of-the-art methods. We achieve a notable margin of superiority over single-stage competitors (*e.g.*, exceeding SeCo [49] by 7.4%, and exceeding PFSR [13] by 8.3%). The results in this table further confirm the superior performance of our method in complex scenarios, thereby proving its overall effectiveness.

Visualizations Figure 3 presents qualitative visualizations of segmentation predictions for single and multiple classes, comparing our method CD-CLIP with other comparable

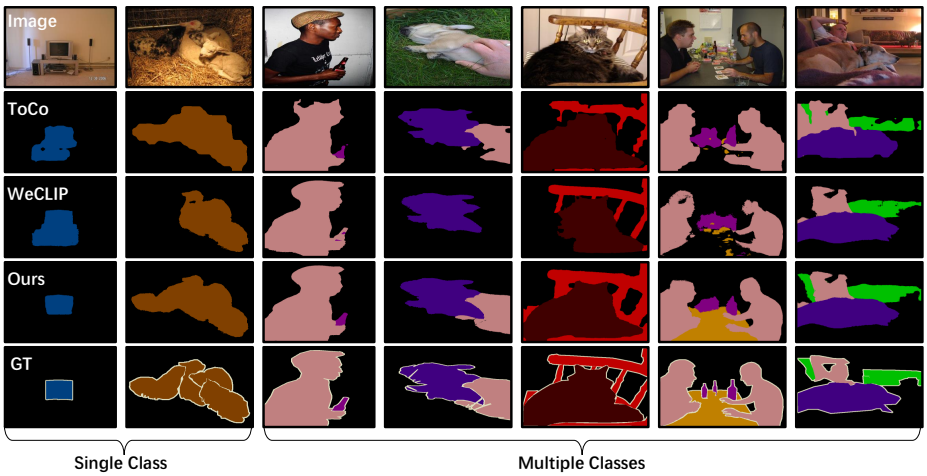


Figure 3. Visualization comparison of final semantic segmentation results between other methods and our method. The experimental dataset is the PASCAL VOC 2012 validation split.

methods to demonstrate its effectiveness. These visualizations distinctly show that our CD-CLIP yields superior segmentation results compared to existing state-of-the-art methods. For multi-class images, CD-CLIP excels at segmenting small objects (*e.g.*, 'bottle') and precisely delineating overlapping instances across different classes (*e.g.*, 'dining table' with 'people' and 'chair' with 'cat') with clear boundaries, which is an important advantage over the other two methods. These notable predictions stem from the capacity for capturing precise patch relationships of our CDA module and the ability to generate accurate super-class boundary regions of our SBE module.

4.3. Ablation Studies

This section presents a comprehensive series of ablation studies designed to assess the efficacy of the Class Distribution Aware (CDA) module, Super-class Boundary Exploration (SBE) module and hyper-parameter selections. Consistent with the single-stage characteristic of our method, we concurrently report performance for both Class Activation Maps (CAMs) and the final segmentation predictions. All experiments are conducted without additional dense CRF post-processing.

Analysis of CD-CLIP In this part, we analyze the effectiveness of our proposed Class Distribution Aware (CDA) and Super-class Boundary Exploration (SBE) module in Table 3 by comparing with the other two excellent affinity modules: Frozen CLIP CAM Refinement module (RFM) [54] and Class-aware Attention-based Affinity (CAA) [27] module under our framework. Among the first three settings (Setting #1, Setting #2, and Setting #3), we test their performance respectively, which proves that our CDA shows the biggest improvement for CAM performance, reaching 80.3% on CAM. Moreover, our CDA module can also be cooperatively used with the other two affinities. When combined with the CAA, the performance increase is limited (Setting #4), as the CAA module contains the regional constraint mask. However, we achieve a significant performance by utilizing CDA and RFM together in Setting #5. Compared to the results in Setting #3 and Setting #6, our SBE module also demonstrates a noticeable improvement.

Table 3. Ablation study of two modules in our method, 'M' denotes the mIoU (%) of CAM performance. 'Seg.' denotes the mIoU (%) of segmentation performance.

#	CAA	RFM	CDA	SBE	M	Seg.
0	No CAM refinement				70.1	68.8
1	✓				73.2	72.8
2		✓			77.4	76.1
3			✓		80.3	80.2
4	✓		✓		76.8	76.4
5		✓	✓		80.1	80.2
6			✓	✓	80.8	81.6

To further substantiate the importance of our CDA module qualitatively, we also present the CAM visualizations across various modules in Figure 4. The result demonstrates the outstanding effect of CDA.

Analysis of Boundary-enhanced Loss $\mathcal{L}_b$ Our Super-class Boundary Exploration (SBE) module employs a specialized boundary-enhanced loss ($\mathcal{L}_b$) to supervise super-class boundary regions. A key aspect of this module is the discriminative selection of supervising images: we focus exclusively on those containing distinct super-classes, guided

Table 4. Ablation study of boundary-enhanced loss $\mathcal{L}_b$ in our SBE module, $\mathbf{C}_\text{p}$ denotes the target class label set, 'Seg.' denotes the mIoU (%) of segmentation performance.

#	$\mathcal{L}_b(\lvert\mathbf{C}_\text{p}\rvert = 1)$	$\mathcal{L}_b(\lvert\mathbf{C}_\text{p}\rvert \geq 2)$	$\mathcal{L}_b$	Seg.
0	w/o $\mathcal{L}_b$			80.2
1	✓			80.4
2		✓		80.8
3			✓	81.6
4	✓	✓		81.4

by class labels. Table 4 provides compelling evidence for this design choice by detailing segmentation prediction performance across different image subsets. Setting #0 establishes our baseline without any auxiliary losses. Applying $\mathcal{L}_b$ solely to single-class images (Setting #1) only improves segmentation prediction performance by 0.2%. This occurs because such images inherently lack the inter-class boundaries $\mathcal{L}_b$ that are designed to supervise, potentially leading to limited boundary improvement. This comprehensive ablation confirms that our meticulous image selection is crucial for the effective supervision and overall success of our $\mathcal{L}_b$ loss in highlighting super-class boundaries.

Analysis of Super-classes Selection Table 5 presents an analysis of how different numbers and sets of super-classes influence performance. We leverage the same super-class set across both the Class Distribution Aware (CDA) module and the Super-class Boundary Exploration (SBE) module, although it plays different roles in each. In the CDA module, it enriches the representation of class distributions, while in the SBE module, it aids in guiding the generation of super-class prototype CAM. To validate the reasonableness of this definition, we designed two additional super-class sets for performance comparison: $\mathbf{D}_{s2} = \{\text{animal, vehicle, indoor things, person}\}$ and $\mathbf{D}_{s3} = \{\text{animal, vehicle, furniture, kitchenware, plant, TV, person}\}$. Although $\mathbf{D}_{s3}$ contains the most classes, some of its super-classes, such as 'kitchenware', may not perfectly align with the target classes. Conversely, the super-classes in $\mathbf{D}_{s2}$ are overly general, which could hinder precise localization. Consequently, we opted for $\mathbf{D}_\text{s}$ as our defined set, as it not only addresses these limitations but also achieves superior performance.

Analysis of Hyper-parameters For the training of our method, we tune the hyper-parameter λ_1 for the boundary-enhanced loss ($\mathcal{L}_b$) within the overall training objective. As shown in Table 6, we conducted several experiments to determine the optimal value for λ_1. After validating λ_1 across the range of 0.1 to 1.0, we find that 0.6 yielded the best results for segmentation prediction.

Table 5. Ablation study of super-classes selection. $\mathbf{D}_\text{s}$, $\mathbf{D}_{s2}$, and $\mathbf{D}_{s3}$ represent different super-class sets, 'M' denotes the mIoU (%) of CAM performance and 'Seg.' denotes the mIoU (%) of segmentation performance.

Set	Number	M	Seg.
$\mathbf{D}_\text{s}$	5	80.8	81.6
$\mathbf{D}_{s2}$	4	80.4	81.1
$\mathbf{D}_{s3}$	7	80.2	81.0

Table 6. Ablation study of λ_1 selection. 'Seg.' denotes the mIoU (%) of segmentation performance.

λ_1	0.1	0.2	0.4	0.6	0.8	1.0
Seg.	80.0	80.1	80.4	81.6	81.2	81.0

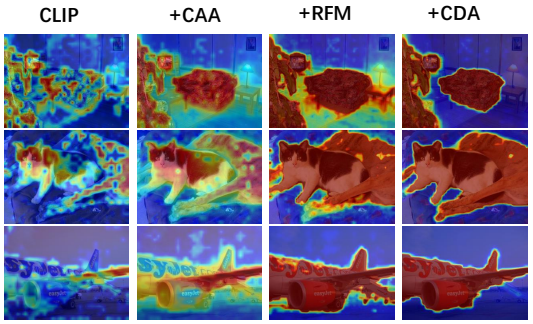


Figure 4. Visualization comparison of CAMs between other modules and our CDA method. The experimental dataset is PASCAL VOC 2012 train split.

5. Conclusion

In this paper, we find class distributions across different classes, instead of target-class activation, contain more valuable semantic information. To fully leverage it, we propose a CD-CLIP method that contains a Class Distribution Aware (CDA) module and refines CAMs by incorporating class distribution relationships. To counteract potential over-activation effects arising from our CDA module, we also introduce a Super-class Boundary Exploration (SBE) module. This SBE module provides accurate supervision for super-class boundary regions, leveraging DINO feature knowledge. Experimental results across various datasets validate the effectiveness of both proposed modules in our CD-CLIP. In future work, we plan to explore relationships for CAM refinement and investigate the benefits of other backbones.

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